



Yen Lin (Darren) Chao

Contact



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Education

Master of Arts in Design Engineering

*Brown University and
Rhode Island School
of Design*

2025-Present

Bachelor of Science in
Electrical Engineering
and Computer Science
*University of
California, Berkeley*

2021-2024

Skills

Design & Fabrication

Autodesk Fusion 360, Adobe Illustrator,
Laser Cutting, 3D Printing/Extrusion, Rapid
Prototyping, Rhino

Languages

Python, Java, C, C++, Go, SQL, RISC-V,
HTML, CSS, Javascript

Cloud & DevOps

Azure IoT Hub, Jenkins, Maven, Git, Jira,
Particle Cloud



Experience

Recycle PLA Researcher

Brown Design Workshop

2025 August - Present

- Architected an end-to-end IoT monitoring system using Particle microcontrollers and sensors to track critical environmental variables (humidity/temperature) during PLA processing.
- Engineered data pipelines by routing real-time sensor data to Azure IoT Hub, automating the flow into a custom SQL database for centralized storage.
- Optimized a closed-loop recycling workflow, managing the transition of PLA waste from pelletizing and grinding to industrial extrusion and heat-pressing.
- Conducted rigorous material testing, including the calibration of laser cutter settings and heat-press parameters (oven/T-shirt press) to ensure structural integrity of upcycled products.
- Led physical product design initiatives, transforming recycled PLA waste into retail-ready consumer goods sold at the Brown University Bookstore.
- Facilitated stakeholder meetings with retail partners and logistics teams to align fabrication capabilities with market demand and pricing strategies.

Telecommunication Private

*Republic of China Army (Taiwan) – 203rd Brigade,
Communications Unit*

2025 Winter

- Operated and maintained military telecommunications systems supporting secure brigade-level communications.
- Performed diagnostics, inspections, and maintenance to ensure operational readiness and reliability.

Software Engineer Intern

Development Bank of Singapore - Taipei

2024 Summer

- Collaborated within an Agile team, following Scrum methodologies with weekly stand-up meetings and sprints.
- Designed and implemented RESTful APIs for the "GiftBox" campaign on the CARD+ app and manipulated XML for form structures and data, directly interacting with the database and processing form inputs.
- Enhanced unit test coverage for legacy APIs to 100%, utilizing mocking and spy frameworks, and Cucumber.
- Deployed and tested APIs using Jenkins, monitored logs for debugging and validation, and managed code using Maven, Git, and Jira.



Coding Projects (see darrenchao.com for design projects)

PACMAN

Modeling Multi-agent Game with ML

February 2024

- Utilized Value Function, Q learning, and Deep Q Network algorithms to enhance PACMAN's gameplay.
- Implemented alpha-beta tree pruning algorithm to efficiently search for and select the optimal path for PACMAN.

REGULUS

Secure File Sharing System

November 2023

- Crafted a multifaceted security infrastructure for the file sharing system, integrating robust user verification mechanisms to authenticate users and stringent file modification protocols to ensure data integrity.
- Employed cryptographic tools including RSA for public-key encryption, HMACs for message authentication, and symmetric key encryption for secure data transmission.
- Established secure communication channels utilizing protocols like TLS/SSL to protect data in transit and mitigate risks associated with interception or eavesdropping.

Build Your Own World

Generating Random Explorable and Interactive Worlds

March 2023

- Architected an algorithm for generating pseudo-random worlds with customizable seeds in Java.
- Implemented a modified Kruskal's algorithm to connect randomly generated rooms with hallways efficiently.
- Currently implemented features include: pseudo-random world generation with connected rooms, the ability to start a new game, load game, or replay last save, light sources, different levels of the game, and locked rooms.